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Deploy HCL DX Compose in Minutes

November 12, 2026



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DX Webinar

An ongoing series of product-focused, hands-on sessions highlighting features and functionality, with an emphasis on demos and visuals.

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Webinar

Deploy HCL DX Compose in Minutes:

Live Demo & Expert Insights!

November 12, 2026 | 5 PM CET



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EuropeSIP DX Expert



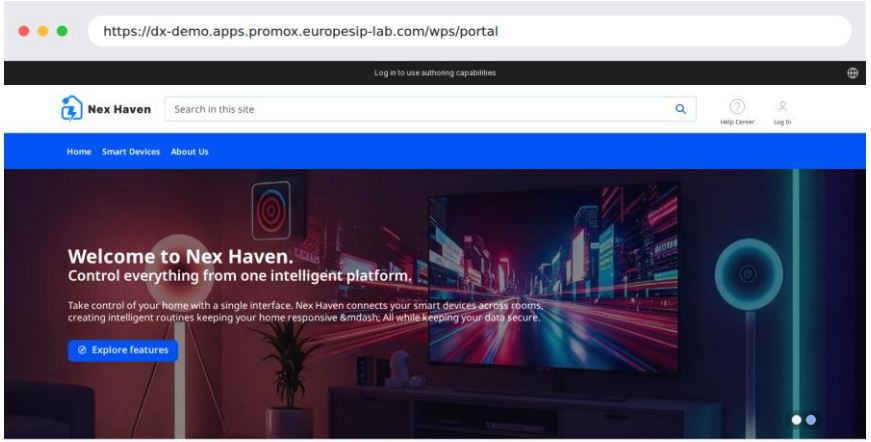
Herbert Hilhorst
HCLSoftware TX
Customer Success Engineer

Deploy HCL DX Compose in Minutes

Ready to take the guesswork out of deploying DX Compose on Kubernetes?

Hands-on session with our trusted Business Partner, EuropeSIP, where we strip away the complexity and build a deployment live—step by step, iteration by iteration.

If you already hold DX licenses, you can deploy DX Compose across your non-production environments today at no extra licensing hurdle!



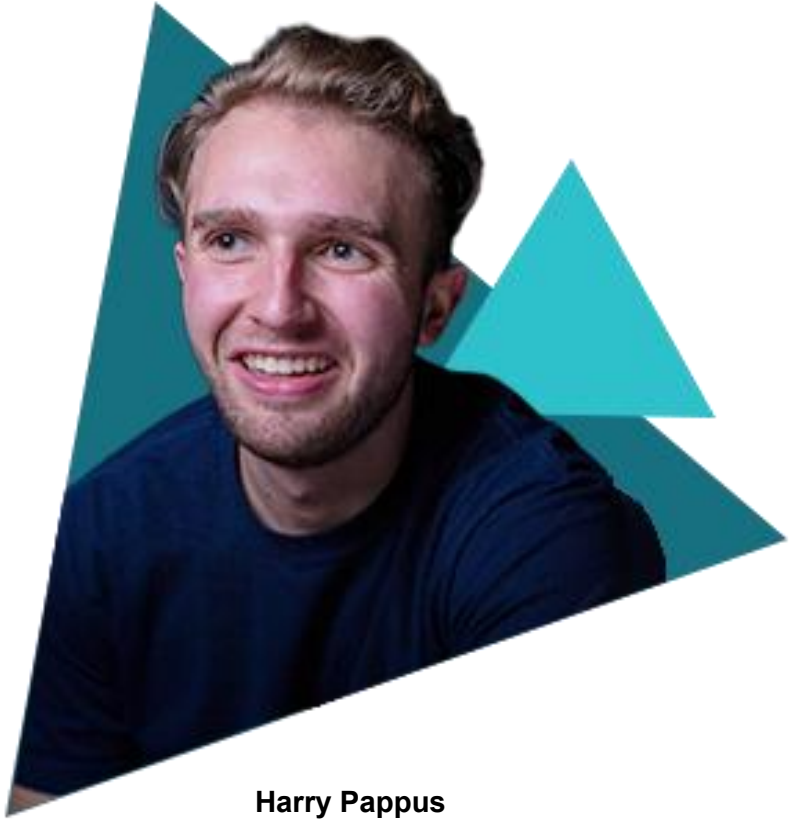
What we build today: HCL DX Compose with the Nex Haven demo site

europesip

Deploy DX Compose in Action

Scenario: Harry Pappus, a DX admin, deploys DX Compose, through a practical, step-by-step incremental approach—building real capabilities one iteration at a time:

1. Base DX with embedded Derby
2. LDAP via Kubernetes Secrets
3. External database
4. AI assistant (IQ)
5. Search with OpenSearch



Harry Pappus

DX Admin, Nex Haven

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Our starting point: the cluster

Kubernetes

- **k3s v1.35.5**, CNCF-certified Kubernetes, managed with Rancher
- **3 control-plane + 3 worker** nodes, Ubuntu 24.04, x86-64
- Workstation: **kubectl** and **Helm 4.3**
- Access to the **HCL Harbor** (hclcr.io) with our CLI secret

DNS, ingress, storage

- **DNS** at our provider: *.apps.promox.europesip-lab.com already points to the cluster
- **Ingress:** ingress-nginx behind MetalLB
- **StorageClasses** (Longhorn 1.12): **dx-deploy-rwo** (ReadWriteOnce) and **dx-deploy-rwx** (ReadWriteMany, for DAM)

Minimum free resources

- **Base DX:** ~5 CPU, ~11 GiB
- **+ LDAP and PostgreSQL:** +0.5 CPU, +1 GiB
- **+ IQ and LiteLLM:** +0.5 CPU, +1.5 GiB
- Start-up peak with limits: ~8 CPU, ~14 GiB, spread over the nodes

The external dependencies

OpenLDAP · pre-installed

- Demo users (andres, macarena…) and groups (**dxadmins**, dxusers)
- openldap.dx-shared:636 (LDAPS)
- Files: **lab-ldap/**

PostgreSQL 18 · pre-installed

- CloudNativePG operator
- **dx_core** prepared for the Web Engine with the HCL table script
- dx-pg-rw.dx-shared:5432 (SSL only)
- Files: **lab-bbdd/**

LiteLLM · live in Lab 4

- Proxy between IQ and the AI models
- OpenRouter → **Claude Sonnet 4.6** / Haiku 4.5
- litellm.dx-shared:4000
- Files: **lab-iq/**

Third-party software, not HCL. OpenLDAP and PostgreSQL were installed before the session, with SSL, as a corporate directory and database would be. LiteLLM, the prerequisite of IQ, is installed live at the start of Lab 4, with IQ. · **dx-demo** = HCL software only · **dx-shared** = these third-party tools.

Today: a production-style deployment

Lab	What we build	⏱ Run time
0	**Prerequisites** Check the preparation · DX namespace	**~2 min**
1	**Base DX** SOPS secrets, HCL charts, DX Compose CF239 on Derby	**~6 min***
2	**LDAP** DX users from OpenLDAP, e-mail sign-in, LDAPS	**~12 min**
3	**External database** Web Engine data: Derby → PostgreSQL over SSL	**~13 min**
4	**AI assistant** LiteLLM, then HCL DX IQ (OpenRouter → Claude)	**~9 min**
5	**Search** OpenSearch and Search V2	**~7 min**
Total run time		**~49 min**

* With the images already in the cluster. The first installation takes 15-25 min, mostly downloading the images.

Looking ahead: towards a GitOps world

Declarative · GitOps-ready

- YAML files instead of imperative commands, Secrets included
- GitOps in large companies: Argo CD or Flux
- Secrets from a manager (HashiCorp Vault, OpenBao, Azure Key Vault) via External Secrets Operator
- Nothing readable in Git: Sealed Secrets or SOPS

SSL, as in a company

- LDAPS on port 636, not plain LDAP on 389
- PostgreSQL only over SSL (sslmode=verify-full)
- Server certificates signed by corporate CAs, trusted by Liberty

Helm 4

- Helm 4 instead of Helm 3
- Small flag differences: --wait=legacy, --force-conflicts
- With Helm 3: --wait, and no --force-conflicts

The HCL documentation shows imperative commands, like kubectl create secret. Here we create the same objects with kubectl apply and YAML files: same result, equally supported. Both ways are valid; we simply prefer YAML.

Prerequisites: the DX namespace

⏱ ~2 min

What we do

- Check what is already in **dx-shared**: OpenLDAP and PostgreSQL
- Create the DX namespace, **dx-demo**: the generic YAML, or the OpenShift one with the annotations HCL requires

Commands · steps 0-1 and 0-2

```
# 0-1 · check the environment
kubectl get nodes
kubectl -n dx-shared get pods             # OpenLDAP, PostgreSQL
kubectl -n dx-shared get clusters.postgresql.cnpg.io
# 0-2 · the DX namespace
cat lab-base/namespace-dx.yaml             # any Kubernetes
cat lab-base/namespace-dx.openshift.yaml   # OpenShift
kubectl apply -f lab-base/namespace-dx.yaml
```

Key values

- **dx-shared**: third-party tools · **dx-demo**: HCL software only
- OpenShift: **openshift.io/sa.scc.*** annotations (restricted-v2)

▶ **Live demo:** show both namespace files and apply one.

📖 HCL documentation (CF239): **DX Compose · Preparation**

Base DX: DX Compose on Derby

⏱ ~6 min*

What we do

- Secrets, encrypted with SOPS: pull secret for hclcr.io (it also signs Helm in to download the HCL charts), **wpsadmin** and a TLS certificate for HAProxy
- Chart **hcl-dx-deployment 2.50.0** (CF239): Web Engine, HAProxy, DAM, Persistence Node
- Check it with kubectl port-forward, then publish it: an Ingress that re-encrypts towards HAProxy
- Result: DX Compose on **Derby**, basic user registry

Commands · steps 1-1 to 1-5

```
# 1-1 · SOPS secrets, HCL charts, TLS for HAProxy
sops -d lab-base/secret-harbor.yaml | kubectl apply -f -
sops -d --extract '[...]' lab-base/secret-harbor.yaml | jq ... \
  | helm registry login hclcr.io -u <user> --password-stdin
helm pull oci://hclcr.io/...   # DX, IQ and Search, -d charts
sops -d -i lab-base/secrets.yaml
kubectl apply -f lab-base/secrets.yaml
openssl req -x509 -newkey rsa:2048 -nodes ... -out lab-base/dx-tls.crt
kubectl -n dx-demo create secret tls dx-tls-cert --cert=... --key=...
# 1-2 · install DX Compose
helm upgrade --install dx-deployment \
     charts/hcl-dx-deployment-2.50.0.tgz -n dx-demo \
     -f lab-base/custom-values.yaml --timeout 30m --wait=legacy
# 1-3 · check DX with kubectl port-forward, before the Ingress
kubectl -n dx-demo port-forward svc/dx-deployment-haproxy 8443:443 &
curl -sk -o /dev/null -w '%{http_code}' https://localhost:8443/wps/portal
# 1-4 · publish it
kubectl apply -f lab-base/ingress-dx.yaml
# 1-5 · check the portal (optional)
```

Key values

- applications: **webEngine** true, core false, openLdap false
- networking.webEngine.host: **dx-demo.apps.promox.europesip-lab.com**
- **dx-deploy-rwo**: Web Engine, Persistence Node, License Manager
- **dx-deploy-rwx**: DAM binaries
- images: hclcr.io + **regcred**

▶ **Live demo:** walk through **lab-base/custom-values.yaml**, then helm install, watch the pods start and log in to /wps/portal as wpsadmin.

📖 **HCL documentation (CF239):** **DX Compose · Installation**

LDAP: DX users from OpenLDAP

⏱ ~12 min

What we do

- DX authenticates against the pre-installed OpenLDAP: the **native LDAP block** of the chart, bind credentials in a Secret
- Administrator role for the **dxadmins** group (XMLAccess)
- Advanced: **configuration overrides** for sign-in by e-mail and the preferred language
- **LDAPS** with the corporate CAs in the Liberty truststore; then plain LDAP is closed

Commands · steps 2-1 to 2-4

```
# 2-1 · the bind secret, then the basic LDAP layer
kubectl apply -f lab-ldap/secrets.yaml
helm upgrade --install dx-deployment ... -f lab-base/custom-values.yaml \
     -f lab-ldap/custom-values.ldap-basic.yaml
# 2-2 · Administrator role for dxadmins (XMLAccess)
curl -sk -u wpsadmin:<password> --data-binary @lab-ldap/admin-role.xml \
     https://<host>/wps/config
# 2-3 · the corporate CAs, then the complete layer (overrides, LDAPS)
kubectl -n dx-demo create secret generic dx-corporate-cas \
     --from-file=internals-ca/corporate-directory-ca.crt \
     --from-file=internals-ca/corporate-database-ca.crt
helm upgrade --install dx-deployment ... -f lab-base/custom-values.yaml \
     -f lab-ldap/custom-values.ldap.yaml
# 2-4 · only LDAPS: OpenLDAP requires TLS (olcSecurity: tls=1)
```

Key values

- ldap.type: **other** · port 389, then **636** and **sslEnabled: true**
- customLdapSecret: **dx-ldap-bind**
- configOverrideFiles: **loginProperty mail**, **wplc-add-property**
- customTruststoreSecrets: **dx-corporate-cas**
- groups: groupOfUniqueNames / uniqueMember

▶ **Live demo:** the basic LDAP layer, the Administrator role and andres; macarena's e-mail fails until the overrides and LDAPS (2-3); then plain LDAP is closed (2-4).

📖 **HCL documentation (CF239):** **DX Compose · LDAP configuration**

External database: Derby → PostgreSQL

⏱ ~13 min

What we do

- The **WebEngine** transfers its data from Derby to **our existing PostgreSQL 18**
- Two secrets with the dbtype / dbdomain properties; JDBC over SSL
- One **helm upgrade** with useExternalDatabase: the chart **transfers** the data
- PostgreSQL needs the table script from HCL (early release): **lab-bbdd/setup-postgres-dx.sql**

Commands · steps 3-1 to 3-3

```
# 3-1 · secrets with the Web Engine database properties
kubectl -n dx-demo create secret generic \
     custom-credentials-webengine-dbtype-secret \
     --from-file=lab-bbdd/dx_dbtype.properties
kubectl -n dx-demo create secret generic \
     custom-credentials-webengine-dbdomain-secret \
     --from-file=lab-bbdd/dx_dbdomain.properties
# 3-2 · transfer Derby -> PostgreSQL
helm upgrade --install dx-deployment ... \
     -f lab-base/custom-values.yaml \
     -f lab-ldap/custom-values.ldap.yaml \
     -f lab-bbdd/custom-values.postgres.yaml
kubectl -n dx-demo logs -f dx-deployment-web-engine-0 \
     -c web-engine
# 3-3 · count the tables (~465)
kubectl -n dx-shared exec dx-pg-1 -c postgres -- psql ...
```

Key values

- useExternalDatabase: **true** · dropDatabaseTables: false
- JDBC: **jdbc:postgresql://dx-pg-rw.dx-shared.svc.cluster.local:5432/dx_core?sslmode=verify-full**
- user **dxuser**, one schema per domain
- Result: 465 tables, over TLS

▶ **Live demo:** walk through **lab-bbdd/custom-values.postgres.yaml** and **dx_dbdomain.properties**, follow the transfer log, count the tables.

📖 HCL documentation (CF239): **DX Compose · Database transfer**

AI assistant: LiteLLM and HCL DX IQ

⏱ ~9 min

What we do

- First **LiteLLM**, the LLM gateway of IQ (not HCL software): **OpenRouter** → **Claude**
- Its secrets: the OpenRouter key, the LiteLLM **master key** and the key IQ uses
- IQ = Integrator + MCP Server (chart **hcl-dx-iq 1.0.3**), deployment key in its values
- One DX **helm upgrade**: HAProxy routes the IQ chat

Commands · steps 4-1 to 4-5

```
# 4-1 · LiteLLM, with its secrets
kubectl apply -f lab-iq/secrets.yaml
helm upgrade --install litellm oci://ghcr.io/berriai/litellm-helm \
     --version 1.106.5 -n dx-shared -f lab-iq/litellm-values.yaml
# 4-2 · ask Claude through LiteLLM (optional)
curl -s localhost:8000/v1/chat/completions -H 'Authorization: ...'
# 4-3 · install IQ (Integrator and MCP Server)
helm upgrade --install dx-iq charts/hcl-dx-iq-1.0.3.tgz -n dx-demo \
     -f lab-iq/custom-iq-values.yaml
# 4-4 · enable IQ in DX
helm upgrade --install dx-deployment ... -f lab-base/... \
     ... -f lab-iq/custom-values.iq.yaml
# 4-5 · ask IQ from the terminal (optional)
```

Key values

- LiteLLM: **litellm.dx-shared:4000**
- iq-general-purpose → **Claude Sonnet 4.6**; iq-summary → **Haiku 4.5**
- Deployment key: in the values (no Secret support yet: SOPS or --set)
- MCP: **enableWcm, enableDam**
- networking.dxIqService: **dx-iq-integrator**

▶ **Live demo:** install **LiteLLM** and ask “ready?”, then walk through **lab-iq/custom-iq-values.yaml** and the IQ button in the DX toolbar: “Which WCM libraries are there?”

📖 **HCL documentation (CF239):** **HCL DX IQ · Deploying LiteLLM** · **HCL DX IQ · Installation**

Search with OpenSearch

⏱ ~7 min

What we do

- **Search V2**: OpenSearch + Search Middleware + File Processor (hcl-dx-search 2.37.0)
- Mutual TLS: a lab CA and **admin / node / client** certificates, one command each, and one secret per certificate
- One DX **helm upgrade** enables Search V2

Commands · steps 5-1 to 5-8

```
# 5-1 · a CA for the lab (in lab-search/certs)
openssl req -x509 -newkey rsa:2048 ... -out root-ca.pem
# 5-2 to 5-4 · admin, node and client certificates, one step each
openssl req -new -newkey rsa:2048 ... -out admin.csr
openssl x509 -req -in admin.csr -CA root-ca.pem ... -out admin.pem
# 5-5 · one secret per certificate (also node, client)
kubectl -n dx-demo create secret generic search-admin-cert \
     --from-file=admin.pem --from-file=admin-key.pem \
     --from-file=root-ca.pem
# 5-6 · install hcl-dx-search
helm upgrade --install dx-search charts/hcl-dx-search-2.37.0.tgz \
     -n dx-demo -f lab-search/custom-search-values.yaml
# 5-7 · enable Search V2 in DX
helm upgrade --install dx-deployment ... -f lab-base/... \
     ... -f lab-iq/... -f lab-search/custom-values.search.yaml
```

Key values

- Secrets: **search-admin-cert, search-node-cert, search-client-cert**
- adminDN: **CN=admin,OU=DX Compose Labs,O=EUROPESIP,C=ES**
- Crawlers: **WCM, JCR, portal and DAM**, every 15 min
- dx-deploy-rwo for the OpenSearch data

▶ **Live demo:** walk through **lab-search/custom-search-values.yaml**, then the Search Center and the search API explorer.

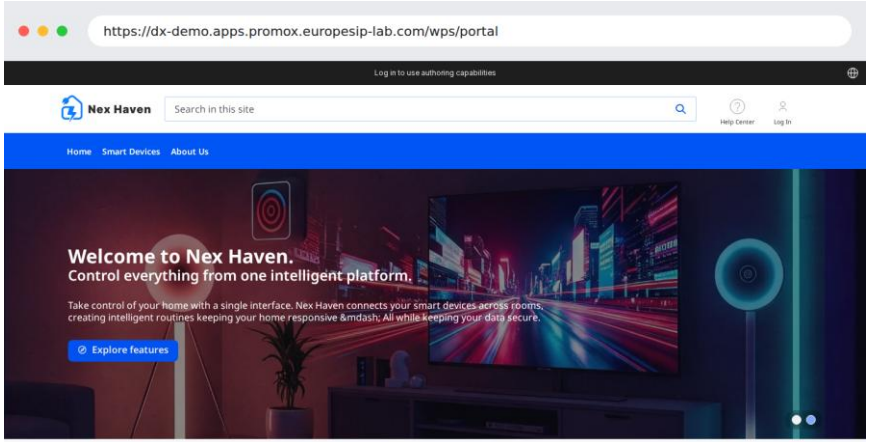
📖 **HCL documentation (CF239):** **DX Compose · Enabling search**

Solution

Deploy DX Compose in Minutes

You can deploy HCL DX Compose, connect to LDAP, transfer to postgreSQL, enable search, IQ and more in no time

1. **Replay this webinar** to review the steps ‘live’ again.
2. **Use the official** documentation details on each step.
3. **Need a hand?** EUROPESIP can help you with any HCL DX installation, in any environment (on premises, cloud or Kubernetes), from a first PoC to a tuned production platform.



Deployed live today: https://dx-demo.apps.promox.europesip-lab.com

Good practices to take home

- Pin the versions of charts and images
- One values file per environment, in Git, reviewed like code
- cert-manager + secrets manager: certificates rotated

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Questions?

You can still add any questions to the Q&A panel, if you haven’t already.

We will address any queries that we don’t get to today in our follow-ups.



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References: the showroom and the installer

The showroom

The files, slides and video of this webinar

github.com/europesip/hcl-dx-compose-showroom

DX Installer Wizard · by EuropeSIP

Generates these files, and a full proof of concept, for your own environment

github.com/europesip/hcl-dx-compose-installer

Disclaimer: this lab and webinar cover a **basic, simple, example installation**. Although it is fully functional, there are many more configuration and **tuning** options not covered here. We always recommend **professional customisation services**, especially for sites with high load expectations.

Free HCL DX resources

HCL Digital Experience · free resources

Marketplace, free resources, documentation, HCLSoftware U and more

hclsw.co/dx-free-resources

Start here: the HCL DX home page, the Help Center and its documentation, HCLSoftware U for training, and the HCLSoftware Marketplace with free assets for DX.

Thank you!

Deploy HCL DX Compose in Minutes

github.com/europesip/hcl-dx-compose-showroom
github.com/europesip/hcl-dx-compose-installer
hclsw.co/dx-free-resources